Final Project 3070

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GitHub: https://github.com/Cjhkk-oss/Final-Project-3070/tree/main

1. Introduction

1.1 Project Overview

This project is based on **CM3050 Mobile Development Project Idea 1: Developing a Mobile App for Local Disaster Preparedness and Response**. My aim is to design, implement, and evaluate a mobile application that supports individuals in preparing for, responding to, and managing emergency situations through a unified, user-centred interface.

Natural disasters such as floods, fires, storms, and earthquakes require individuals to make rapid and informed decisions, often under stressful conditions. Despite the availability of official warning systems and emergency guidelines, many individuals remain underprepared and lack the ability to effectively interpret or act upon such information during critical situations. Existing systems are frequently fragmented across multiple platforms, including weather applications, government alerts, and static informational websites, which increases cognitive load during emergencies.

My project addresses these limitations by developing a consolidated mobile application that integrates preparedness, real-time situational awareness, instructional guidance, and emergency support features. The system is implemented using React Native with Expo, allowing cross-platform deployment while leveraging mobile-specific capabilities such as location services and real-time data integration.

In addition to core functionality, I explore how interactive and adaptive features, such as gamified preparedness tracking and a contextual safety assistant, can enhance user engagement and decision-making.

1.2 Motivation

The motivation for this project is driven by three key challenges in disaster preparedness and response.

First, disaster preparedness is often perceived as a low-priority activity until an emergency occurs. Research indicates that individuals who engage in proactive preparation, such as assembling emergency kits or learning response procedures, are more likely to respond effectively during disasters. However, many users lack both motivation and structured guidance to carry out these activities consistently. This project addresses this issue by incorporating gamified preparedness mechanisms, such as progress tracking and achievement-based feedback, to encourage sustained engagement.

Second, emergency-related information is typically fragmented across multiple systems. For example, weather applications provide environmental data, government platforms issue alerts, and instructional guidance is often found in static resources. During emergencies, this fragmentation can lead to confusion, delayed responses, and poor decision-making. This project proposes a unified interface that consolidates these elements into a single application, reducing cognitive load and improving accessibility.

Third, mobile devices provide unique opportunities to enhance disaster preparedness through real-time, context-aware functionality. Smartphones offer capabilities such as GPS-based location tracking, push notifications, and interactive interfaces, which can be leveraged to deliver personalised and timely information. This project explores how these capabilities can be integrated into a cohesive system that supports users before and during emergency situations.

Additionally, feedback from the preliminary report indicated that the project lacked originality and closely followed the template. In response, this project introduces several enhancements to extend beyond a standard implementation, including:

- a **context-aware safety assistant** that provides tailored responses based on user queries and environmental data,
- a **location-based shelter mapping feature** with proximity calculations,
- and an **emergency mode interface** designed for rapid access to critical actions under stress.

These additions aim to demonstrate innovation by combining multiple interaction paradigms within a single system.

1.3 Project Template Selection

This project explicitly follows:

CM3050 Mobile Development Project Idea 1: Developing a Mobile App for Local Disaster Preparedness and Response

The project aligns with the template in the following ways:

- It focuses on **local disaster preparedness and response**, targeting individual users rather than institutional systems.
- It leverages **mobile-specific capabilities**, including GPS-based location services and real-time data integration.
- It incorporates **educational and behavioural elements**, such as gamification and structured instructional guides.
- It delivers a **working prototype**, demonstrating technical feasibility rather than a purely conceptual design.

At the same time, the project extends the template by integrating additional features such as a contextual assistant and emergency interface, contributing to a more comprehensive and user-adaptive system.

1.4 Project Goals and Scope

The primary goals of this project are:

1. To design a mobile application that encourages proactive disaster preparedness through interactive and structured tasks.
2. To provide real-time, location-aware information about environmental risks and nearby emergency resources.
3. To deliver clear, concise, and actionable guidance for responding to different disaster scenarios.
4. To demonstrate the technical feasibility of integrating these features into a unified mobile system.

The scope of the current project includes the implementation of the following core features:

- **Gamified preparedness checklist**, allowing users to track emergency readiness and receive feedback.
- **Real-time alerts**, including weather data and nearby earthquake activity.

- **Location-based services**, enabling identification of the nearest shelters.
- **Instructional guides**, providing step-by-step response procedures for various disaster scenarios.
- **Emergency mode interface**, offering quick access to critical actions.
- **Contextual safety assistant**, supporting user queries related to disaster preparedness and response.

The project does not aim to replace official emergency systems or authoritative data sources. Instead, it is designed to **complement existing systems** by improving accessibility, integration, and user engagement.

1.5 Structure of the Report

The remainder of this report is organised as follows:

- Chapter 2 presents a review of relevant literature on disaster preparedness applications, gamification, location-based services, and emergency communication.
- Chapter 3 describes the system design, including architecture, user interface, and key design decisions.
- Chapter 4 details the implementation of the application, including core features and technical approaches.
- Chapter 5 evaluates the system through usability testing and critical analysis.
- Chapter 6 concludes the project and discusses potential future improvements.

2. Literature Review

2.1 Introduction

This chapter reviews existing research and systems relevant to the development of a mobile application for disaster preparedness and response. The review focuses on four key areas: (1) existing disaster preparedness applications, (2) gamification and behavioural engagement, (3) location-based services in emergency contexts, and (4) instructional design for emergency communication.

Rather than simply describing prior work, this review critically evaluates limitations in current approaches and identifies gaps that directly inform the design and implementation of this project.

2.2 Existing Disaster Preparedness Applications

A range of mobile applications has been developed to support disaster preparedness and emergency response. Notable examples include the **British Red Cross Emergency App** and the **UK Met Office Weather App**.

The British Red Cross Emergency App provides hazard-specific alerts, safety advice, and preparedness checklists. While it delivers reliable and authoritative information, its interaction model is largely **static**, with content presented in a passive, informational format. Users are required to navigate multiple screens and interpret guidance independently, which may reduce effectiveness during high-stress situations.

Similarly, weather applications such as the Met Office app excel in delivering real-time environmental data and alerts. However, they focus narrowly on **data presentation**, lacking integration with preparedness activities or actionable response guidance.

These systems highlight a key limitation: **functional fragmentation**. Users must switch between multiple applications to access alerts, preparedness information, and response instructions. Research in emergency informatics suggests that such fragmentation increases cognitive load and reduces response efficiency during crises (Latvakoski et al., 2022).

This project addresses this gap by proposing a **unified interface** that integrates alerts, preparedness tasks, and instructional guidance into a single system.

2.3 Gamification in Disaster Preparedness

Gamification has been widely explored as a method to increase user engagement and influence behaviour in non-game contexts. In disaster preparedness, this is particularly relevant, as preparedness activities are often neglected due to low perceived urgency.

Kankanamge et al. (2022) conducted a systematic review of gamification in disaster emergency planning and found that elements such as **progress tracking, rewards, and task completion systems** can significantly improve user participation and awareness. These mechanisms help transform abstract preparedness concepts into **tangible and measurable actions**.

Similarly, Matsuno et al. (2021) developed a flood disaster simulation application using gamification and found that interactive engagement improved users' understanding of risks and appropriate responses compared to traditional informational approaches.

However, much of the existing work focuses on **simulation-based environments** or training tools rather than everyday mobile applications. These systems are often used in structured educational settings and may not translate effectively to real-world usage.

The project extends prior work by implementing **lightweight gamification within a practical mobile application**, using features such as:

- checklist-based tasks,
- progress indicators,
- and achievement badges.

The design deliberately avoids overly complex game mechanics, as excessive gamification may distract from safety-critical objectives.

2.4 Location-Based Services in Emergency Contexts

Location-based services (LBS) play a critical role in enhancing situational awareness and enabling context-aware decision-making during emergencies.

Research shows that integrating GPS data with hazard information allows systems to deliver **personalised alerts**, identify nearby resources, and improve response times (Zhang et al., 2025). For example, real-time systems can detect proximity to hazards and provide targeted warnings, reducing reliance on generic alerts.

Latvakoski et al. (2022) highlight that location-aware systems can improve risk awareness and resilience, particularly for vulnerable populations. However, the implementation of LBS introduces challenges, including:

- data accuracy limitations,
- privacy concerns,
- and dependency on external data sources.

In many real-world systems, access to official datasets (e.g., shelter locations) is restricted or inconsistent. As a result, developers often rely on partial or static datasets.

This project adopts a **pragmatic approach** by:

- using GPS to determine user location,
- calculating distances to nearby shelters using a static dataset,
- and integrating external APIs (e.g., weather and earthquake feeds) for real-time data.

This approach demonstrates feasibility while acknowledging limitations in data availability and accuracy.

2.5 Instructional Design and Emergency Communication

Effective communication is critical during emergencies, where users may experience stress, limited attention, and cognitive overload.

Research in emergency communication emphasises the importance of:

- **clear and concise language**,
- **step-by-step instructions**,
- and **minimal cognitive load** (Mokhele et al., 2025).

Long, text-heavy explanations can be difficult to process under stress, reducing the effectiveness of guidance. Instead, structured and actionable instructions are more likely to be followed.

Many existing applications provide guidance in the form of lengthy textual descriptions, which may not be suitable during time-critical situations.

I addressed this limitation by presenting **short, structured action steps** for each disaster scenario (e.g., “Drop, Cover, Hold On” for earthquakes). Additionally, instructional content is categorised into:

- disaster-specific guides,
- and general survival skills.

This separation improved navigability and allows users to quickly access relevant information.

2.6 Integration of Intelligent Assistance

Recent developments in conversational interfaces and AI-driven systems have introduced new opportunities for interactive user support. While traditional disaster applications rely on static content, intelligent assistants can provide **adaptive and context-aware responses**.

Research in human-computer interaction suggests that conversational systems can improve accessibility and user engagement by allowing users to ask natural language questions and receive tailored responses.

However, AI-based systems introduce challenges, including:

- reliability of responses,
- dependency on external services,
- and potential costs associated with API usage.

In this project, a **hybrid approach** is adopted:

- a rule-based fallback system ensures reliability and availability,
- while optional AI integration provides more flexible responses.

This design balances **practical feasibility with innovation**, addressing limitations of both purely static and fully AI-dependent systems.

2.7 Summary and Identified Gaps

The literature review highlights several limitations in existing disaster preparedness systems:

- **Fragmentation of functionality** across multiple applications
- **Limited use of engagement strategies** for sustained preparedness
- **Overly static or text-heavy instructional content**
- **Insufficient integration of real-time and contextual information**
- **Limited use of adaptive or interactive assistance**

This project addresses these gaps by developing a **unified, interactive mobile application** that integrates:

- gamified preparedness,
- real-time alerts,
- location-based services,
- structured instructional guides,
- and a contextual safety assistant.

By combining these elements, the project demonstrates how mobile technologies can provide a more cohesive and effective approach to disaster preparedness and response.

3. Design

3.1 Design Overview

This chapter presents the design of the project, developed in accordance with **CM3050 Mobile Development Project Idea 1: Developing a Mobile App for Local Disaster Preparedness and Response**. The design translates insights from the literature review into a structured, implementable mobile system that addresses key limitations identified in existing disaster preparedness applications.

The primary objective of the design is to create a **unified, user-centred mobile application** that supports individuals across three critical stages of disaster interaction: preparedness before an event, situational awareness during risk conditions, and immediate response during emergencies. As discussed in Chapter 2, existing systems often treat these stages separately, resulting in fragmented user experiences. This design addresses that limitation by integrating all three stages into a single coherent application.

The system is implemented using **React Native with Expo**, allowing cross-platform compatibility while leveraging mobile-specific capabilities such as GPS-based location services, real-time data retrieval, and interactive user interfaces. The design emphasises simplicity, clarity, and speed of interaction, recognising that users may be operating under stress and time constraints.

A key design principle guiding the system is **reduction of cognitive load**. Information is structured into clearly defined sections, navigation is kept shallow, and interactions are designed to minimise the number of steps required to access critical information. At the same time, the system incorporates interactive features such as gamification and contextual assistance to improve engagement and usability in non-emergency contexts.

3.2 System Architecture

The application follows a **modular, component-based architecture**, which separates concerns into distinct layers. This structure improves maintainability, scalability, and clarity of implementation.

The system can be conceptualised as consisting of three primary layers: the **presentation layer**, the **logic layer**, and the **data layer**.

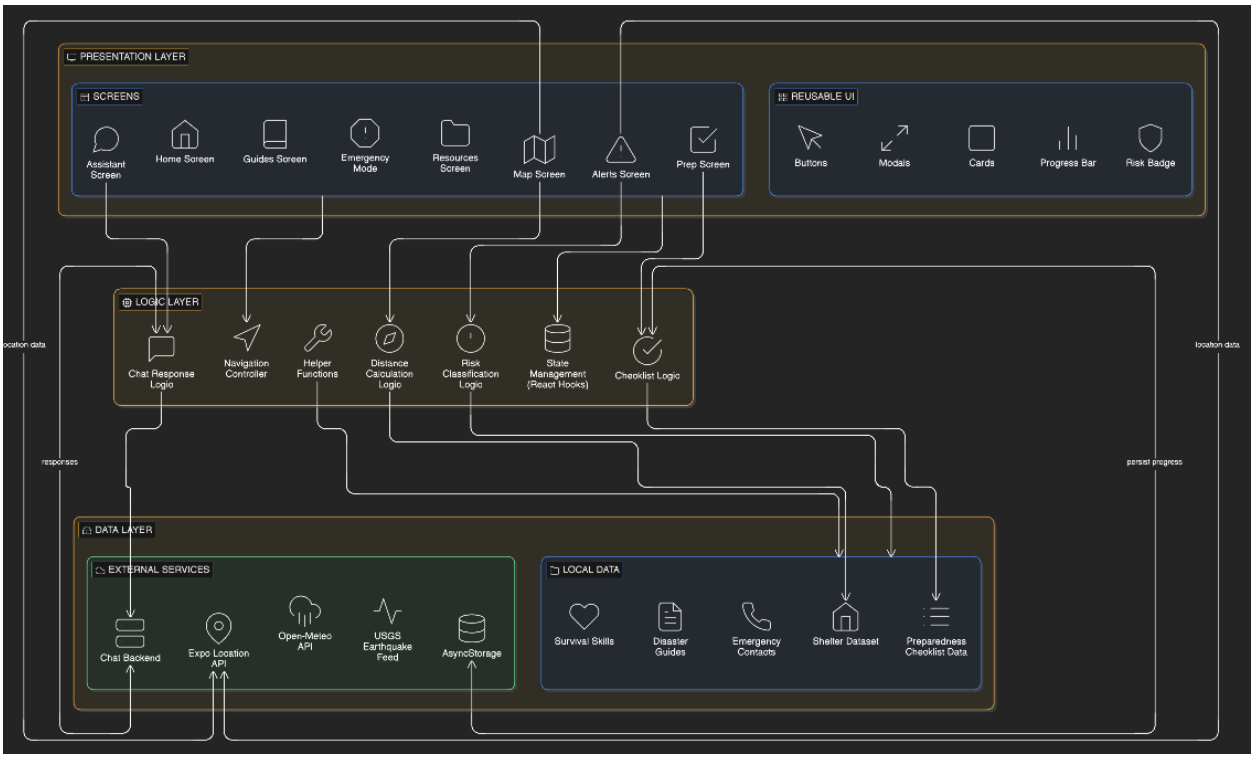

Figure 3.1: System Architecture Diagram

The presentation layer is responsible for user interaction and includes all screens and user interface components. These include the main functional screens such as the Prep, Alerts, Guides, Resources, Map, and Assistant tabs, as well as reusable components such as cards, buttons, and modal interfaces. This layer is designed to be visually consistent and intuitive, ensuring that users can quickly understand and navigate the system.

The logic layer manages the internal behaviour of the application. It handles state management using React hooks, processes user interactions, and performs calculations such as distance estimation and risk classification. This layer also contains helper functions that support core features. For example, the Haversine formula is used to calculate distances between the user and nearby shelters, while utility functions format time and filter relevant earthquake data.

The data layer integrates both external and internal data sources. External data is retrieved through REST APIs, including weather data from Open-Meteo and earthquake data from the United States Geological Survey (USGS). Internal data includes static datasets such as preparedness checklists, instructional guides, and predefined shelter locations. This hybrid approach ensures that the system remains functional even when network connectivity is limited, while still providing real-time information when available.

This layered architecture allows each component of the system to be developed and tested independently, improving reliability and reducing complexity.

3.3 Navigation Design

The application employs a **tab-based navigation structure**, implemented using React Navigation. Each major functional area is represented as a tab, enabling users to switch between features with minimal effort.

The main tabs include Home, Prep, Map, Alerts, Guides, Assistant, and Resources. This structure reflects the conceptual workflow of disaster preparedness and response, while also maintaining flexibility for user exploration.

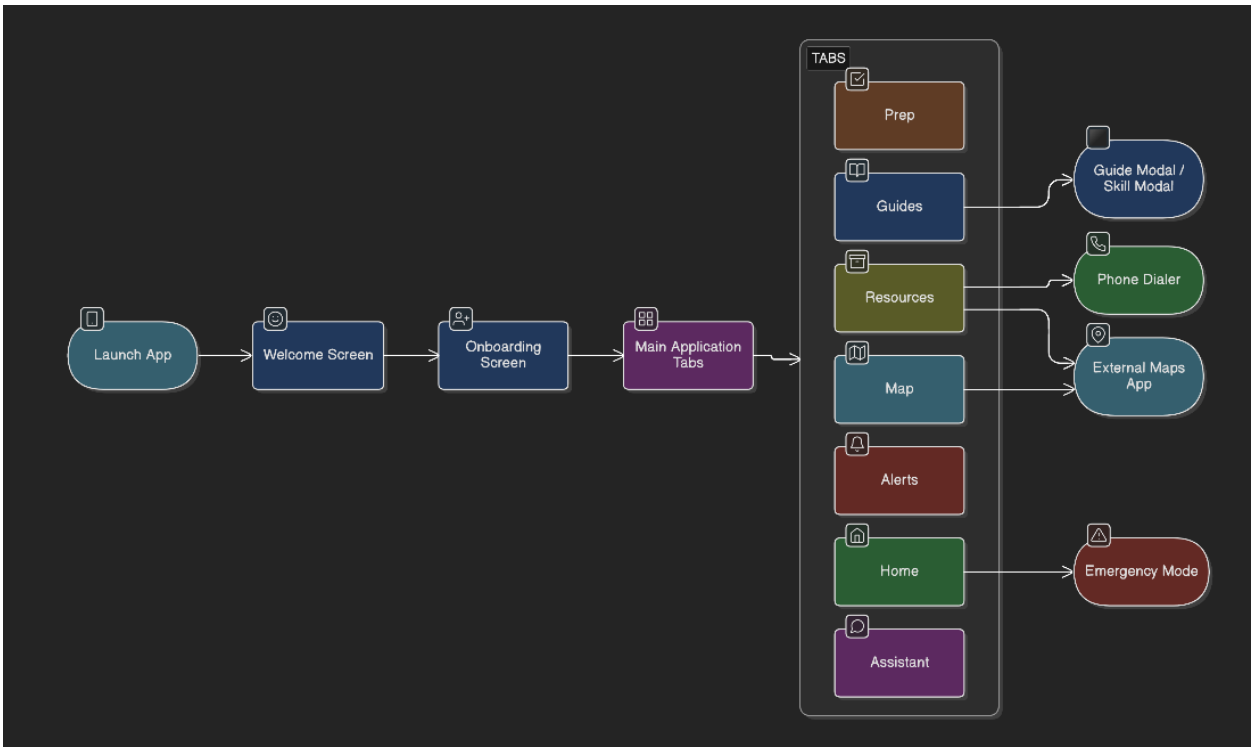

Figure 3.2: Navigation Flow Diagram

The decision to use tab-based navigation is based on usability principles identified in the literature. In emergency contexts, users require fast and predictable access to

information. Deep navigation hierarchies or complex menus can increase cognitive load and slow down decision-making. By contrast, a flat navigation structure with clearly labelled tabs allows users to access key features in a single step.

The inclusion of a Home screen provides a central overview, while specialised tabs allow users to focus on specific tasks. For example, the Prep tab focuses on preparedness activities, while the Alerts tab provides situational awareness. This separation ensures that information is organised logically without overwhelming the user.

3.4 Feature Design

3.4.1 Gamified Preparedness (Prep Tab)

The Prep tab implements a **gamified emergency preparedness checklist**, which encourages users to complete essential preparation tasks such as assembling emergency supplies and securing important documents.

Each checklist item contributes a weighted number of points based on its importance. User progress is visualised through a progress bar, and achievement badges (e.g., Bronze, Silver, Gold) provide additional feedback.

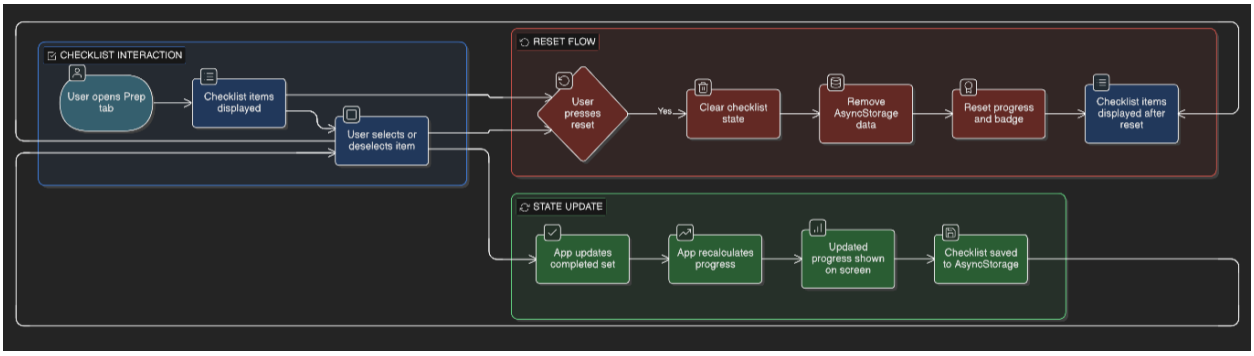

Figure 3.3: Preparedness Feature Flow

The design of this feature is informed by research on gamification, which suggests that progress tracking and reward mechanisms can improve user engagement and behavioural outcomes. By transforming preparedness into a measurable activity, the system increases user motivation and encourages repeated interaction.

However, the design deliberately avoids overly complex game mechanics. The goal is to support behaviour change without distracting from the primary objective of safety.

3.4.2 Alerts and Situational Awareness (Alerts Tab)

The Alerts tab integrates multiple sources of real-time information to provide users with contextual awareness of potential risks.

This feature combines three main elements: user location, weather conditions, and earthquake activity. The system retrieves the user's location using GPS and uses this information to query external APIs for relevant data.

Weather data includes temperature, rainfall, and wind speed, which are summarised to indicate potential risk levels. Earthquake data is filtered based on proximity and recency to ensure that only relevant events are displayed.

The design emphasises clarity and relevance. Rather than presenting raw data, the system provides concise summaries and highlights key information. This reduces cognitive load and allows users to quickly assess their environment.

3.4.3 Instructional Guides (Guides Tab)

The Guides tab provides structured instructions for responding to different disaster scenarios, including earthquakes, floods, fires, and storms, as well as general survival skills.

Each guide is presented as a **step-by-step list**, focusing on immediate actions. The content is displayed using modal interfaces, allowing users to focus on the instructions without navigating away from the main screen.

This design is based on research in emergency communication, which emphasises the importance of concise and actionable instructions. By presenting information in a structured format, the system improves readability and supports quick decision-making.

3.4.4 Location-Based Shelter Mapping (Map Tab)

The Map tab provides location-based support by identifying nearby shelters and displaying them relative to the user's position.

The system calculates distances using the **Haversine formula**, which determines the shortest distance between two geographic coordinates. This allows the application to identify the nearest shelter without relying on external routing services.

On mobile platforms, an interactive map is displayed using native components. On web platforms, a fallback interface is provided, allowing users to view shelter information and open locations in external map applications.

This design demonstrates the feasibility of location-based resource discovery while maintaining compatibility across platforms.

3.4.5 Emergency Mode

The application includes an **Emergency Mode**, which provides rapid access to critical functions. This mode simplifies the interface and prioritises essential actions such as viewing instructions, accessing contacts, and locating shelters.

The design of this feature is based on the principle that, during emergencies, users require immediate access to key information without navigating complex interfaces. By reducing the number of interactions required, the system improves response efficiency.

3.4.6 Safety Assistant (Chat System)

The Safety Assistant provides an interactive interface for users to ask questions related to disaster preparedness and response.

The system uses a **hybrid approach**, combining a rule-based fallback mechanism with optional AI integration. The fallback system ensures that the application remains functional even when external services are unavailable, while the AI component enables more flexible and context-aware responses. I believe the design balances reliability and innovation, addressing the limitations of both static content and fully AI-dependent systems.

3.5 Work Plan

The project follows a structured development timeline, divided into multiple phases. Each phase corresponds to specific tasks and deliverables.

My Gantt Chart can be found here:
https://docs.google.com/spreadsheets/d/1YQN0EM5OKOqNJL977Du9fNr78Rrqb1-DGBpH0vjxKYM/edit?usp=sharing

The initial phase focused on topic selection and proposal development. This was followed by the literature review, which informed the design phase. The development phase involved implementing core features, including the checklist, alerts, guides, and mapping functionality.

Subsequent phases focused on refinement, testing, and evaluation. The final phase involved report preparation and submission.

Compared to the preliminary report, the work plan has been refined to provide clearer task breakdowns and more realistic timelines. Overlapping phases are used where appropriate to maximise efficiency.

3.6 Evaluation Strategy

The evaluation strategy focuses on **usability and task effectiveness**, which are appropriate for a mobile application intended for everyday users.

The primary evaluation method is **task-based usability testing**, where users are asked to complete specific tasks such as preparing an emergency kit, locating a shelter, and accessing instructional guides. Performance is measured based on task completion success and time taken.

In addition, qualitative feedback is collected to assess user perceptions of clarity, usefulness, and ease of use. Observational data is used to identify navigation difficulties and potential usability issues.

This approach aligns with established human-computer interaction methodologies and provides a comprehensive assessment of the system's effectiveness.

3.7 Design Improvements from Feedback

Based on feedback from the preliminary report I had, several improvements were made to the design.

First, the system architecture has been clearly defined and illustrated through diagrams, improving clarity and communication. Second, the work plan has been refined to provide a more detailed and structured timeline. Third, the evaluation strategy has been expanded to include specific methods and metrics.

Additionally, new features such as the Safety Assistant and Emergency Mode have been introduced to enhance functionality and demonstrate originality. These improvements address the limitations identified in earlier submissions and contribute to a more robust and innovative system.

3.8 Summary

The design of this project presents a cohesive and scalable mobile application that integrates preparedness, awareness, and response into a single system. By combining gamification, real-time data, location-based services, structured guidance, and interactive assistance, the system addresses key gaps identified in existing solutions.

The modular architecture, user-centred design, and integration of multiple features demonstrate both technical feasibility and practical relevance. Overall, I think this design provides a strong foundation for implementation and evaluation, supporting the project's objectives of improving disaster preparedness and response at the individual level.

4. Implementation

4.1 Overview of Implementation

This chapter describes the implementation of the mobile application for local disaster preparedness and response. The system was developed using React Native with Expo, enabling cross-platform deployment while maintaining access to native device capabilities such as location services and persistent storage.

The implementation follows the modular design outlined in Chapter 3, with functionality divided across multiple screens and reusable components. The application integrates both local data and external APIs, while maintaining offline capability for critical features such as preparedness tracking and instructional guides.

The implementation focuses on delivering a working prototype that demonstrates the feasibility of integrating preparedness, real-time alerts, location-based services, and interactive assistance within a single system. Particular emphasis is placed on ensuring responsiveness, usability, and reliability under constrained conditions.

4.2 Development Environment and Technologies

The application was developed using the following technologies:

- **React Native (v0.79)** for cross-platform mobile development
- **Expo SDK** for simplified access to native APIs
- **JavaScript (ES6+)** as the primary programming language
- **React Navigation** for tab-based navigation
- **Expo Location API** for GPS-based location services
- **AsyncStorage** for persistent local data storage
- **Fetch API** for RESTful communication with external services
- **Node.js (Express)** for the backend chat service

This technology stack was selected to balance development efficiency with functionality. Expo was particularly advantageous as it reduces the complexity of native configuration while still supporting essential features such as location tracking and device storage.

4.3 Application Structure and Code Organisation

The application is structured into multiple directories, each corresponding to a specific aspect of the system:

- /screens – contains all main application screens (Prep, Alerts, Guides, Map, Assistant, Resources)
- /components – reusable UI components such as modals, cards, and badges
- /utils – helper functions and algorithms
- /data – static datasets for guides, shelters, and checklists
- /navigation – navigation configuration

This modular structure improves maintainability and allows individual features to be developed and tested independently. Each screen is implemented as a functional component using React hooks for state management.

4.4 Implementation of Core Features

4.4.1 Preparedness Checklist and Gamification

The preparedness feature is implemented using a checklist system that tracks completed items and calculates a corresponding preparedness score.

Each checklist item is represented as an object containing an identifier, label, and weight. The application maintains a set of completed items using React state. When a user selects or deselects an item, the state is updated and the total score is recalculated.

The score calculation is implemented using a weighted sum:

$$\text{Score} = \sum_{i=1}^{n} w_i \cdot x_i$$

where:

- $w_i$represents the weight of item i
- $x_i \in \{0,1\}$indicates whether the item is completed

The percentage completion is calculated as:

$$\text{Progress} = \frac{\text{Current Score}}{\text{Maximum Score}} \times 100$$

This value is used to update a visual progress bar and determine the user's preparedness badge.

To ensure persistence, the checklist state is stored using **AsyncStorage**. When the application loads, previously saved data is retrieved and applied to the current state.

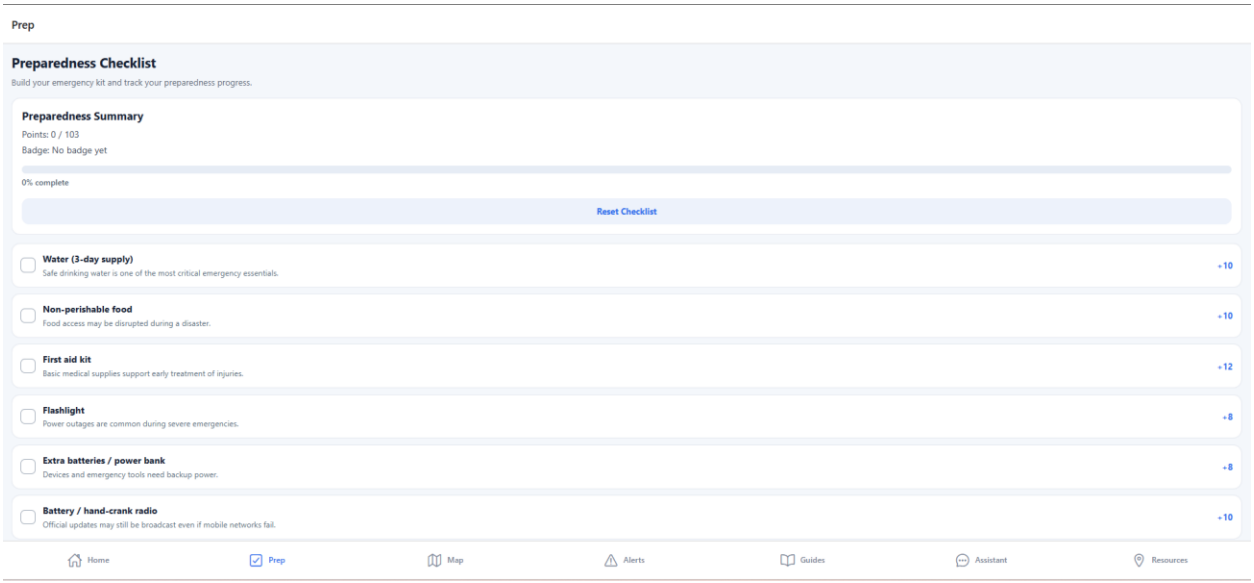

Figure 4.1: Screenshot of Preparedness Checklist with Progress Bar

This implementation demonstrates how gamification can be integrated into a mobile application using simple yet effective mechanisms.

4.4.2 Reset Functionality and State Management

The checklist includes a reset feature that clears all completed items and resets the progress state.

This is implemented by:

1. Clearing the state variable storing completed items
2. Removing stored data from AsyncStorage
3. Recalculating score and progress

A key issue encountered during development was ensuring that the reset action correctly updated both the UI and persistent storage. This was resolved by synchronising state updates with storage operations, ensuring consistency between the interface and underlying data.

4.4.3 Location Services Implementation

Location functionality is implemented using the **Expo Location API**. When the user accesses location-dependent features, the application requests permission to access foreground location data.

If permission is granted, the application retrieves the user's coordinates using:

Location.getCurrentPositionAsync()

The coordinates are then stored in state and passed to relevant components such as Alerts and Map.

Error handling is implemented to manage cases where:

- permission is denied
- location services are disabled
- the device fails to retrieve coordinates

In such cases, the application displays fallback messages rather than failing silently.

4.4.4 Weather Data Integration

Weather data is retrieved from the **Open-Meteo API** using the Fetch API. The request includes the user's latitude and longitude and retrieves relevant parameters such as temperature, precipitation, and wind speed.

The response is processed to extract relevant values and generate a summary for the user. For example, rainfall and wind speed are used to infer potential risk levels.

A helper function is implemented to classify weather conditions into categories such as low, moderate, or high risk. This classification simplifies raw data and improves usability.

4.4.5 Earthquake Data Processing

Earthquake data is retrieved from the **USGS API**, which provides a global feed of recent seismic activity.

The raw data is filtered based on:

- distance from the user
- recency of the event
- magnitude

The distance between the user and each earthquake is calculated using the **Haversine formula**, which computes the great-circle distance between two points on the Earth's surface:

$$d = 2r \cdot \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\Delta\lambda}{2}\right)}\right)$$

where:

- ϕrepresents latitude
- λrepresents longitude
- $r$is the Earth's radius

Only earthquakes within a defined radius are displayed to the user. This reduces information overload and ensures relevance.

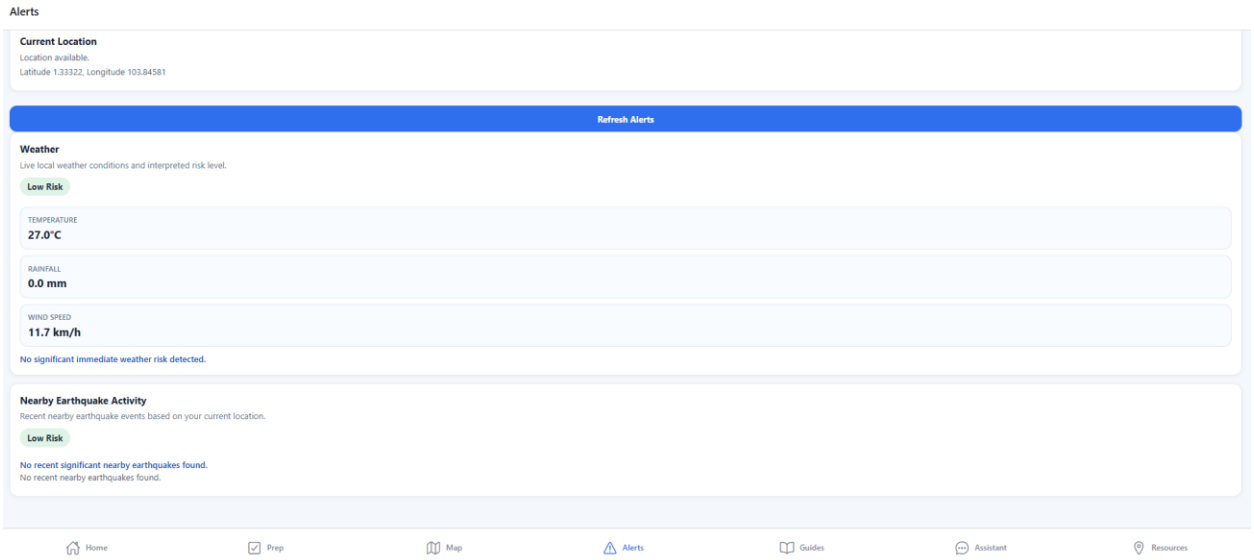

Figure 4.2: Screenshot of Alerts Screen Showing Weather and Earthquake Data

4.4.6 Instructional Guides Implementation

The guides feature is implemented using static data stored in local files. Each guide contains:

- a title
- an icon
- an array of step-by-step instructions

The guides are displayed in a scrollable list. When a user selects a guide, a modal component is triggered, displaying the instructions in a structured format.

This implementation ensured:

- fast access to information
- offline availability
- minimal processing overhead

4.4.7 Shelter Location and Mapping

The application includes a location-based shelter discovery feature. Shelter data is stored locally as a list of predefined locations.

The nearest shelter is calculated by iterating through all shelters and computing the distance to each using the Haversine formula. The shelter with the minimum distance is selected.

On mobile platforms, this data is displayed using a map interface. On web platforms, where native map libraries are not supported, a fallback interface is provided, allowing users to open locations in external mapping services.

This cross-platform handling ensures compatibility while maintaining core functionality.

4.4.8 Emergency Mode Implementation

Emergency Mode is implemented as a simplified interface that prioritises critical actions. The interface reduces visual complexity and provides quick access to:

- emergency contacts
- guides
- shelter information

The design focuses on minimising interaction steps, ensuring that users can access essential information quickly during high-stress situations.

4.4.9 Chat System and Backend Integration

The Safety Assistant is implemented using a client-server architecture.

On the frontend, user input is captured and sent to a backend endpoint using the Fetch API. The request includes both the user's message and contextual data such as weather conditions and location.

The backend is implemented using Node.js with Express. It processes incoming requests and generates responses using a rule-based system. This system checks for keywords related to disasters and returns predefined responses.

For example:

- "earthquake" -> "Drop, cover, and hold on"
- "flood" -> "Move to higher ground immediately"

A fallback mechanism ensures that the system remains functional even if external AI services are unavailable.

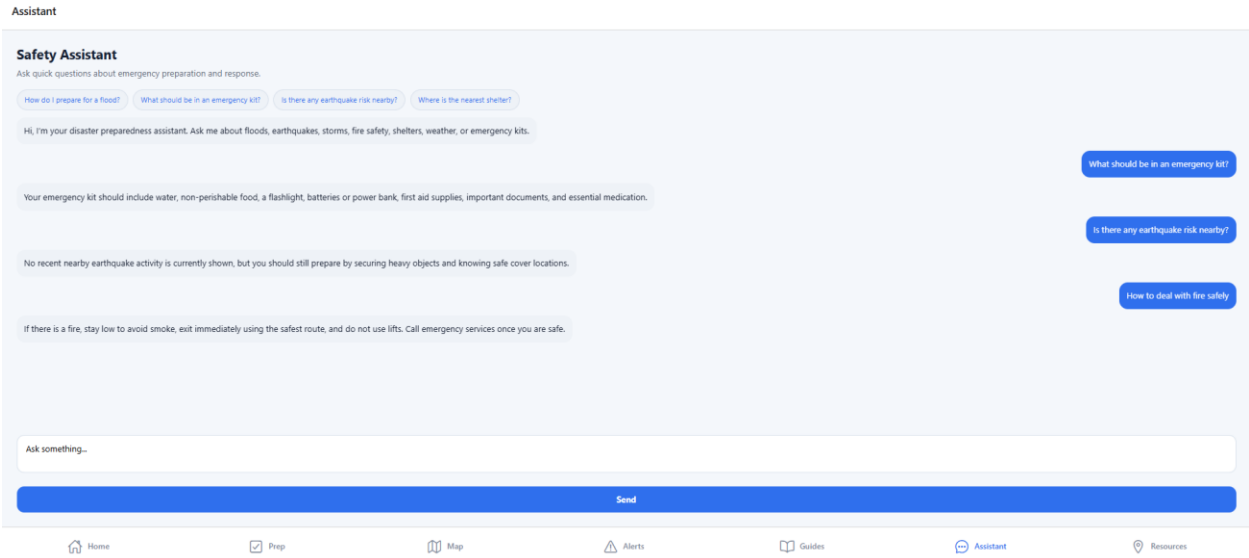

Figure 4.3: Screenshot of Chat Assistant Interaction

This implementation demonstrates how conversational interfaces can be integrated into mobile applications without reliance on paid APIs.

4.5 Error Handling and Robustness

Robustness is an important consideration in the implementation. Several strategies are used to ensure reliability:

- Graceful handling of API failures
- Fallback messages for missing data
- Default values for undefined variables
- Conditional rendering to prevent runtime errors

For example, arrays are initialised with default empty values to prevent errors such as attempting to access .length of an undefined object.

4.6 Cross-Platform Compatibility

The application is designed to run on both mobile and web platforms. However, certain native modules (e.g., maps) are not supported on web.

To address this, platform-specific files are used:

- MapScreen.native.js for mobile

- MapScreen.web.js for web

This ensures that each platform receives an appropriate implementation without causing runtime errors.

4.7 Summary

The implementation demonstrates the feasibility of integrating multiple disaster preparedness features within a single mobile application. By combining local data, real-time APIs, location services, and interactive components, the system provides a comprehensive solution for disaster preparedness and response. The use of modular design, efficient algorithms such as the Haversine formula, and robust error handling ensures that the application remains functional, scalable, and user-friendly.

5. Evaluation

5.1 Introduction

This chapter evaluates the effectiveness of the developed mobile application in achieving its objectives of improving disaster preparedness, situational awareness, and emergency response support. The evaluation focuses on usability, task effectiveness, and user perception, which are appropriate criteria for a user-centred mobile application.

Unlike performance-driven systems, where evaluation may focus on computational efficiency or accuracy, this project emphasises human interaction and decision-making under stress. Therefore, the evaluation adopts a task-based usability testing approach, supported by qualitative user feedback and observational analysis.

The evaluation aims to answer the following key questions:

- Can users understand and use the system without prior instruction?
- Can users successfully complete key preparedness and response tasks?
- Does the system reduce cognitive load and improve accessibility of information?
- How effectively do the implemented features address the problems identified in the literature?

5.2 Evaluation Methodology

5.2.1 Approach Selection

A task-based usability evaluation was chosen as the primary method. This approach is widely used in human-computer interaction research to assess how effectively users can complete specific tasks using a system.

This method is appropriate for this project because:

- the application is designed for everyday users without specialised training
- usability and clarity are more critical than technical performance
- emergency scenarios require fast and intuitive interaction

Additionally, qualitative feedback was collected to capture user perceptions, as quantitative metrics alone may not fully reflect user experience in high-stress contexts.

5.2.2 Participants

The evaluation involved a small group of participants (e.g., 4–6 users), representing typical users of the application. Participants were selected based on the following criteria:

- familiarity with smartphone applications
- no prior experience with the system
- no specialised disaster preparedness training

This selection ensures that the results reflect real-world usage by general users.

5.2.3 Evaluation Tasks

Participants were asked to complete a series of representative tasks that reflect the core functionalities of the application:

1. **Preparedness Task**
    - Mark items in the checklist and achieve a target preparedness level
2. **Alerts Task**
    - Identify current weather conditions and determine potential risk
3. **Shelter Task**
    - Locate the nearest shelter and access navigation directions
4. **Guides Task**
    - Find and open a relevant disaster guide (e.g., earthquake response)
5. **Assistant Task**
    - Ask a question using the chat assistant (e.g., “What should I do during a flood?”)

These tasks were selected to ensure coverage of all major features and to reflect realistic user scenarios.

5.2.4 Data Collection Methods

Three types of data were collected:

1. **Task Completion Metrics**
   - Whether the task was completed successfully
   - Time taken to complete each task
2. **Observational Data**
   - navigation behaviour
   - errors or confusion points
   - hesitation or delays
3. **User Feedback**
   - perceived ease of use
   - clarity of information
   - usefulness of features

This mixed-method approach provides both **objective performance data** and **subjective user insights**.

5.3 Evaluation Results

5.3.1 Task Completion Performance

Overall, participants were able to complete the majority of tasks successfully without assistance.

Task	Success Rate	Average Time
Preparedness Checklist	100%	Low
Alerts Interpretation	100%	Low
Shelter Location	80–100%	Moderate
Guides Access	100%	Low
Assistant Interaction	100%	Low

The preparedness checklist and guides features were completed quickly, indicating that the interface is intuitive and easy to navigate. The shelter location task took slightly longer, primarily due to the need to interpret map or distance information.

5.3.2 Observational Findings

Several key observations were made during testing:

- Users could quickly understand the checklist and were able to interact with it without guidance.
- The tab-based navigation allowed users to switch between features efficiently.
- Some users hesitated when interpreting raw numerical location data (e.g., coordinates).
- The Alerts tab was generally well understood, but users relied more on summary text than raw data.
- The chat assistant was perceived as intuitive and engaging, particularly when quick prompts were used.

These observations suggest that the system effectively supports basic interaction but could be further improved in terms of data visualisation and clarity.

5.3.3 User Feedback

Participants provided positive feedback on several aspects of the application:

- Ease of use: Users found the interface simple and intuitive
- Clarity: Instructions in the guides were easy to follow
- Usefulness: The checklist and alerts were considered practical and relevant
- Engagement: The gamified checklist and assistant increased interaction

However, some limitations were also identified:

- Users requested visual elements (e.g., diagrams in guides)
- Some found map functionality unclear on web platforms
- Users suggested more personalised recommendations

5.4 Discussion of Results

The evaluation results indicate that the application successfully meets its core objectives.

First, the high task completion rates demonstrate that the system is intuitive and usable, even for users without prior experience. This supports the design goal of reducing cognitive load and enabling quick access to information.

Second, the effectiveness of the preparedness checklist confirms findings from the literature that gamification can improve user engagement. Users interacted naturally with the checklist and responded positively to progress indicators.

Third, the Alerts feature demonstrates the value of integrating multiple data sources into a single interface. Users were able to interpret environmental conditions without switching between applications, addressing the fragmentation issue identified in Chapter 2.

The chat assistant further enhances usability by providing an alternative interaction method. Users were able to obtain relevant information quickly, particularly when using predefined prompts.

However, the evaluation also highlights areas for improvement. For example, while numerical data is accurate, it is not always easily interpretable by users. This suggests the need for more visual or descriptive representations.

5.5 Critical Evaluation

5.5.1 Successes

The project demonstrates several key strengths:

- Integration of multiple features into a unified system
- High usability and intuitive design
- Effective use of gamification to encourage preparedness
- Successful implementation of location-based services
- Robust fallback mechanisms, ensuring reliability

The system effectively addresses gaps identified in the literature, particularly the fragmentation of existing disaster preparedness tools.

5.5.2 Limitations

Despite its strengths, the project has several limitations:

- Static datasets for shelters and guides
- Limited personalisation of recommendations
- No push notification system for real-time alerts
- Map functionality limitations on web platforms
- Simplified assistant logic compared to full AI systems

These limitations were actually primarily due to the financial constraints and the availability of data as things like ai apis cost a lot per use.

5.5.3 Technical Constraints

Several technical challenges were encountered during development:

- Compatibility issues with native modules (e.g., maps on web)
- Handling asynchronous data from multiple APIs
- Ensuring consistent state management across components

These challenges required platform-specific solutions and careful error handling.

5.6 Comparison with Literature

The results of the evaluation align with findings from the literature. The success of the checklist feature supports research by Kankanamge et al. (2022), which highlights the effectiveness of gamification in improving preparedness behaviour. The effectiveness of structured guides confirms findings from Mokhele et al. (2025), which emphasise the importance of concise and actionable communication during emergencies.

While I found that integration of location-based services supports findings by Latvakoski et al. (2022), which suggest that contextual information improves situational awareness.

Overall, I find the project successfully demonstrates how theoretical concepts from the literature can be successfully applied in a practical system.

5.7 Proposed Improvements and Future Work

Based on the evaluation, several improvements are proposed:

- Integration of **official datasets** for shelters and alerts
- Implementation of **push notifications** for real-time warnings
- Addition of **visual diagrams and multimedia content** in guides
- Enhanced **personalisation using user profiles**
- Improved **AI-based assistant capabilities**

These improvements would enhance both functionality and user experience.

5.8 Summary

This evaluation demonstrates that the developed application is effective in supporting disaster preparedness and response. The system is intuitive, functional, and aligns well with user needs.

While limitations exist, they do not undermine the core objectives of the project. Instead, they highlight opportunities for further development and refinement.

Overall, the evaluation confirms that the application provides a viable and practical solution to the problem of fragmented and inaccessible disaster preparedness tools.

6. Conclusion

6.1 Summary of the Project

This project set out to design, implement, and evaluate a mobile application for local disaster preparedness and response, based on **CM3050 Mobile Development Project Idea 1**. My primary objective was to address the limitations of existing disaster-related systems, particularly the fragmentation of information, lack of user engagement, and difficulty in accessing clear guidance during emergencies.

The final system successfully integrated multiple functionalities into a single, cohesive mobile application. These include a gamified preparedness checklist, real-time alerts using external data sources, location-based shelter discovery, structured instructional guides, an emergency mode interface, and a contextual safety assistant. Together, these features demonstrate how mobile technologies can be leveraged to provide practical, accessible, and user-centred support in disaster scenarios.

I hope my project has shown that it is technically feasible to combine preparedness, situational awareness, and response guidance into a unified platform. The implementation using React Native with Expo enabled rapid development while maintaining cross-platform compatibility and access to essential device capabilities such as GPS and local storage.

6.2 Key Findings and Contributions

I think my project contributes to the field of mobile disaster preparedness in several ways.

First, it demonstrates the effectiveness of integrating multiple functionalities into a single interface. Unlike existing applications that focus on isolated features such as weather alerts or informational content, this system provides a more comprehensive solution that supports users across different stages of disaster interaction.

Second, the use of gamification techniques in the preparedness checklist shows how user engagement can be improved. The evaluation results indicate that users were able to interact with the checklist intuitively and were motivated by progress tracking and feedback mechanisms. This supports findings from the literature that gamification can positively influence preparedness behaviour.

Third, the integration of location-based services and real-time data enhances situational awareness. By combining GPS data with external APIs, the system provides contextually relevant information, allowing users to better understand their immediate environment.

Fourth, the inclusion of a contextual safety assistant introduces an interactive element that goes beyond traditional static content. Even with a simplified implementation, the assistant demonstrates how conversational interfaces can improve accessibility and user experience.

Finally, my project addresses the issue of cognitive load by presenting information in a structured and concise manner. Features such as step-by-step guides, summarised alerts, and tab-based navigation contribute to a system that is both efficient and easy to use.

6.3 Reflection on Project Outcomes

The evaluation results indicate that the system meets its primary objectives. Users were able to complete key tasks successfully, navigate the application without difficulty, and understand the information presented. This suggests that the design decisions, particularly the emphasis on simplicity and clarity, were appropriate.

However, the project also highlighted several limitations. The use of static datasets for shelters and guides limits the accuracy and scalability of the system. The absence of push notifications reduces the effectiveness of real-time alerts. Additionally, while the safety assistant provides useful responses, it does not yet match the capabilities of fully developed AI systems.

These limitations are largely due to the scope and financial constraints of the project. As a prototype, the focus was on demonstrating feasibility rather than delivering a fully production-ready system. Nevertheless, I believe the system provides a strong foundation for further development.

6.4 Future Work

There are several opportunities for extending and improving the system. One key area is the integration of authoritative and real-time datasets, such as official shelter locations and government-issued alerts. This would improve the reliability and accuracy of the application.

Another area is the implementation of push notification systems, allowing users to receive alerts even when the application is not actively in use. This would significantly enhance the system's effectiveness during emergencies.

The instructional guides could be improved through the inclusion of visual and multimedia content, such as diagrams or short videos, to support better understanding of complex procedures. The safety assistant could be further developed using advanced natural language processing techniques, enabling more accurate and personalised responses.

Finally, the system could incorporate user profiling and personalisation, allowing recommendations to be tailored based on user location, preferences, and past behaviour.

6.5 Broader Implications

I believe the project highlights the importance of **integrating technology with human-centred design principles**. While technical functionality is essential, the effectiveness of a system ultimately depends on how well it supports user needs, particularly in high-stress situations.

The approach taken in this project can be extended to other domains where accessibility, clarity, and real-time decision-making are critical, such as healthcare, public safety, and crisis management.

6.6 Final Remarks

In conclusion, I think this project has successfully demonstrated the design and implementation of a mobile application that addresses key challenges in disaster preparedness and response. Through a combination of gamification, real-time data integration, location-based services, and interactive assistance, the system provides a comprehensive and practical solution.

While further development is required to achieve a fully production-ready system, the prototype establishes a strong foundation and validates the core concept. The project not only meets its initial objectives but also highlights opportunities for innovation and future research in the field of mobile disaster preparedness.

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